

DISCRETE MATHEMATICAL STRUCTURES

(DMS)

Complete Handwritten-Style Exam Notes

- Unit I – Logic, Proofs & Set Theory
- Unit II – Functions, Sequences & Recurrence Relations
- Unit III – Counting Techniques & Relations
- Unit IV – Graph Theory & Trees

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Full Paper Name: **DMS — Discrete Mathematical Structures**

"Notes designed to be easy to understand and exam-ready — with definitions, diagrams, visualizations, solved examples, and quick-revision boxes for every topic and sub-topic."

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QUICK REVISION

Formula & Identity Cheat-Sheet

Important Exam Questions — Unit-wise

UNIT I Logic, Proofs & Set Theory

1.1 Propositional Logic

Definition: A **Proposition** (or statement) is a declarative sentence that is either **True (T)** or **False (F)**, but not both. **Propositional Logic** is the branch of logic that deals with propositions and the logical operators that combine them.

Example — Propositions vs Non-Propositions

'Rajendra University is in Balangir' $\rightarrow$ Proposition (True)

' $2 + 2 = 5$ ' $\rightarrow$ Proposition (False)

'What time is it?' $\rightarrow$ NOT a proposition (a question, no truth value)

' $x + 2 = 5$ ' $\rightarrow$ NOT a proposition alone (truth depends on x — this is a predicate, see Section 1.3)

Logical Connectives (Operators)

Connective	Symbol	Name	Example (p = 'It rains', q = 'Ground is wet')
NOT	$\neg p$	Negation	$\neg p$ = 'It does not rain'
AND	$p \wedge q$	Conjunction	$p \wedge q$ = 'It rains AND ground is wet'
OR	$p \vee q$	Disjunction	$p \vee q$ = 'It rains OR ground is wet'
IF...THEN	$p \rightarrow q$	Implication/Conditional	'If it rains, then ground is wet'
IFF	$p \leftrightarrow q$	Biconditional	'It rains if and only if ground is wet'
XOR	$p \oplus q$	Exclusive OR	True when exactly one of p, q is true

Truth Tables of Basic Connectives

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Understanding the Conditional $p \rightarrow q$

Definition: $p \rightarrow q$ ('if p then q ') is FALSE only when p is True and q is False. p is called the **hypothesis/antecedent**, q is called the **conclusion/consequent**.

Related Conditional	Form	Example (p: ' $n > 2$ ', q: ' $n^2 > 4$ ')
Converse	$q \rightarrow p$	If $n^2 > 4$ then $n > 2$
Inverse	$\neg p \rightarrow \neg q$	If $n \leq 2$ then $n^2 \leq 4$
Contrapositive	$\neg q \rightarrow \neg p$	If $n^2 \leq 4$ then $n \leq 2$

★ **Exam Tip:** The contrapositive ($\neg q \rightarrow \neg p$) is **always logically equivalent** to the original conditional ($p \rightarrow q$) — this fact is used heavily in proof by contrapositive. The converse and inverse are NOT equivalent to the original.

1.2 Propositional Equivalences

Definition: Two propositions are **Logically Equivalent** (written $p \equiv q$) if they have identical truth values in every possible row of their truth table. A proposition that is always True is a **Tautology**; one that is always False is a **Contradiction**; one that is sometimes True and sometimes False is a **Contingency**.

Example — Tautology Example

$p \vee \neg p$ is **always True** $\rightarrow$ Tautology (Law of Excluded Middle)
 $p \wedge \neg p$ is **always False** $\rightarrow$ Contradiction

Important Laws of Logical Equivalence

Law Name	Equivalence
Identity Laws	$p \wedge T \equiv p$ $p \vee F \equiv p$
Domination Laws	$p \vee T \equiv T$ $p \wedge F \equiv F$
Idempotent Laws	$p \vee p \equiv p$ $p \wedge p \equiv p$
Double Negation	$\neg(\neg p) \equiv p$
Commutative Laws	$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$
Associative Laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive Laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan's Laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption Laws	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
Negation Laws	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$
Conditional Identity	$p \rightarrow q \equiv \neg p \vee q$

Example — Proving Equivalence using Laws

Show $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$
 $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ [De Morgan]
 $\equiv \neg p \wedge (p \vee \neg q)$ [De Morgan again]
 $\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ [Distributive]
 $\equiv F \vee (\neg p \wedge \neg q)$ [Negation Law]
 $\equiv \neg p \wedge \neg q$ [Identity Law] ✓ Proved

★ **Exam Tip:** De Morgan's Laws are the single most-tested equivalence — practice applying them to negate compound statements with AND/OR quickly.

1.3 Predicates & Quantifiers

Definition: A **Predicate** is a statement containing one or more variables that becomes a proposition once specific values are substituted for the variables. **P(x): 'x > 5'** is a predicate — true or false depends on x.

Quantifiers

Quantifier	Symbol	Meaning	Example
Universal Quantifier	$\forall x P(x)$	P(x) is true for ALL x in the domain	$\forall x (x^2 \geq 0)$ — true for all real x
Existential Quantifier	$\exists x P(x)$	P(x) is true for AT LEAST ONE x	$\exists x (x^2 = 4)$ — true since x=2 works

Negating Quantified Statements

De Morgan's Laws for Quantifiers

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

In words: 'Not all x satisfy P' means 'some x does NOT satisfy P', and 'No x satisfies P' means 'all x fail to satisfy P'.

Example — Negating a Quantified Statement

Statement: $\forall x (x \text{ is a student} \rightarrow x \text{ has an ID card})$

Negation: $\exists x (x \text{ is a student} \wedge x \text{ does NOT have an ID card})$

1.4 Nested Quantifiers

Definition: Nested Quantifiers occur when one quantifier appears within the scope of another. The **order** of different quantifiers ($\forall$ vs $\exists$) changes the meaning.

Example — Nested Quantifiers — Order Matters

Let $P(x,y)$: ' $x + y = 0$ ' over integers.

$\forall x \exists y P(x,y) \rightarrow$ 'For every x , there EXISTS a y such that $x+y=0$ ' $\rightarrow$ **TRUE** ($y = -x$ always exists)

$\exists y \forall x P(x,y) \rightarrow$ 'There EXISTS a single y that works for EVERY x ' $\rightarrow$ **FALSE** (no single y satisfies all x)

★ **Exam Tip:** $\forall x \exists y$ and $\exists y \forall x$ are NOT the same — this is one of the most common exam trap questions. Always translate nested quantifiers into plain English first.

1.5 Rules of Inference

Definition: Rules of Inference are logical forms used to derive valid conclusions from a set of given premises — the foundation of formal mathematical proof.

Rule Name	Form	Example
Modus Ponens	$p, p \rightarrow q \therefore q$	It rains, If it rains ground is wet $\therefore$ ground is wet
Modus Tollens	$\neg q, p \rightarrow q \therefore \neg p$	Ground not wet, If rain then wet $\therefore$ it didn't rain
Hypothetical Syllogism	$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$	Chain of implications
Disjunctive Syllogism	$p \vee q, \neg p \therefore q$	It's Mon or Tue, not Mon $\therefore$ it's Tue
Addition	$p \therefore p \vee q$	p is true, so ' p or anything' is true
Simplification	$p \wedge q \therefore p$	If both true, each individually is true
Conjunction	$p, q \therefore p \wedge q$	Combine two known truths
Resolution	$p \vee q, \neg p \vee r \therefore q \vee r$	Used in automated theorem proving

Example — Applying Rules of Inference

Premise 1: If I study, I pass. ($p \rightarrow q$)

Premise 2: I study. (p)

Conclusion (Modus Ponens): **I pass. (q)**

★ **Exam Tip:** Modus Ponens and Modus Tollens are asked almost every semester — memorise their exact forms and don't confuse them (Modus Tollens negates the CONCLUSION to negate the hypothesis).

1.6 Introduction to Proofs

Common Proof Techniques

Technique	Approach
Direct Proof	Assume p is true, use logical steps to directly show q is true
Proof by Contrapositive	Prove $\neg q \rightarrow \neg p$ instead of $p \rightarrow q$ (logically equivalent)

Technique	Approach
Proof by Contradiction	Assume the negation of the conclusion is true, derive a contradiction
Proof by Induction	Prove base case $P(1)$, then prove $P(k) \rightarrow P(k+1)$ for all k
Proof by Cases	Split into exhaustive cases and prove each case separately
Proof by Counterexample	Disprove a universal statement by finding ONE case where it fails

Example — Direct Proof Example

Prove: 'If n is an even integer, then n^2 is even.'

Let n be even $\rightarrow n = 2k$ for some integer k .

$n^2 = (2k)^2 = 4k^2 = 2(2k^2) \rightarrow$ this is $2 \times (\text{integer}) \rightarrow n^2$ is even. **QED**

Example — Proof by Contradiction Example

Prove: ' $\sqrt{2}$ is irrational.'

Assume the opposite: $\sqrt{2} = a/b$ in lowest terms (a, b integers, no common factor).

Then $2b^2 = a^2$, so a^2 is even $\rightarrow a$ is even $\rightarrow a = 2c$.

Substituting: $2b^2 = 4c^2 \rightarrow b^2 = 2c^2 \rightarrow b$ is also even.

But then a and b share a factor of 2 — contradicts 'lowest terms'. **Hence $\sqrt{2}$ is irrational. QED**

★ **Exam Tip:** Proof by contradiction and direct proof are the two most commonly examined proof techniques — practice the ' $\sqrt{2}$ is irrational' and 'sum of two odd numbers is even' style questions.

1.7 Sets — Venn Diagrams, Subsets & Size of a Set

Definition: A **Set** is an unordered collection of distinct objects, called **elements** or **members**. Sets are usually denoted by capital letters, e.g. $A = \{1, 2, 3, 4\}$.

Ways of Describing a Set

Method	Example
Roster/Tabular Form	$A = \{1, 2, 3, 4, 5\}$
Set-Builder Form	$A = \{x \mid x \text{ is a natural number, } x \leq 5\}$
Venn Diagram	Pictorial representation using circles inside a rectangle (universe)

Membership & Special Sets

Notation	Meaning
$x \in A$	x is an element/member of set A
$x \notin A$	x is NOT an element of set A

Notation	Meaning
$\emptyset$ or $\{ \}$	Empty/Null Set — contains no elements
U	Universal Set — the set of all elements under consideration
$ A $ or $n(A)$	Cardinality — the number of elements in set A

Subsets

Definition: Set A is a **Subset** of B (written $A \subseteq B$) if every element of A is also an element of B . A is a **Proper Subset** ($A \subset B$) if $A \subseteq B$ and $A \neq B$ (B has at least one extra element).

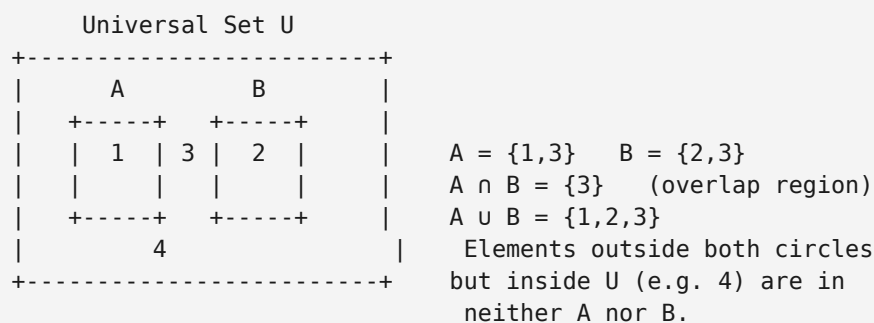
Example — Subsets

$A = \{1,2\}$ $B = \{1,2,3\} \rightarrow A \subseteq B$ and $A \subset B$ (proper)

Every set is a subset of itself: $A \subseteq A$. The empty set is a subset of every set: $\emptyset \subseteq A$.

Venn Diagrams

Fig 1.9 — Venn Diagram of Two Sets



Size (Cardinality) of a Set

- **Finite Set:** Has a countable number of elements, e.g. $\{1,2,3\}$.
- **Infinite Set:** Has unlimited elements, e.g. the set of natural numbers $\mathbb{N}$.
- **Cardinality** of A = number of elements = $|A|$. Example: $A = \{a,b,c\} \rightarrow |A| = 3$.

1.8 Power Sets & Cartesian Products

Power Set

Definition: The **Power Set** of A , written $P(A)$, is the set of **all possible subsets** of A , including the empty set and A itself. If $|A| = n$, then $|P(A)| = 2^n$.

Example — Power Set

$A = \{1, 2\} \rightarrow P(A) = \{ \emptyset, \{1\}, \{2\}, \{1,2\} \} \rightarrow |P(A)| = 2^2 = 4$

Cartesian Product

Definition: The **Cartesian Product** $A \times B$ is the set of all **ordered pairs** (a, b) where $a \in A$ and $b \in B$. $|A \times B| = |A| \times |B|$.

Example — Cartesian Product

$$A = \{1, 2\} \quad B = \{x, y\}$$

$$A \times B = \{(1,x), (1,y), (2,x), (2,y)\} \rightarrow |A \times B| = 2 \times 2 = 4$$

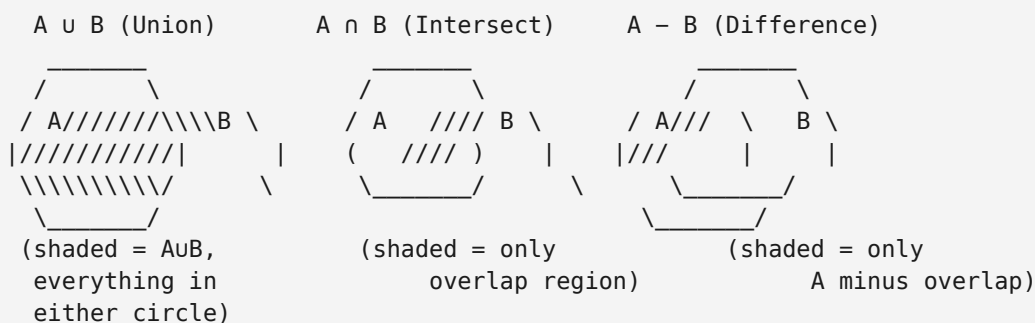
Note: $A \times B \neq B \times A$ in general (order matters in the pairs).

★ **Exam Tip:** Power set size formula $|P(A)| = 2^n$ is a guaranteed 1-2 mark numerical — also remember that a Cartesian product is NOT commutative.

1.9 Set Operations

Operation	Symbol	Definition
Union	$A \cup B$	Elements in A OR B (or both)
Intersection	$A \cap B$	Elements in BOTH A and B
Difference	$A - B$	Elements in A but NOT in B
Complement	A' or A^c	Elements in U but NOT in A
Symmetric Difference	$A \oplus B$	Elements in A or B but not both $= (A-B) \cup (B-A)$

Fig 1.10 — Set Operations (Venn Diagrams)



Set Identities (Laws of Set Algebra)

Law	Identity
Commutative	$A \cup B = B \cup A \quad A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
De Morgan's Laws	$(A \cup B)' = A' \cap B' \quad (A \cap B)' = A' \cup B'$
Identity	$A \cup \emptyset = A \quad A \cap U = A$
Complement	$A \cup A' = U \quad A \cap A' = \emptyset$

Principle of Inclusion-Exclusion

Inclusion-Exclusion Formula

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For three sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Example — Inclusion-Exclusion

In a class of 50 students, 30 like Tea, 25 like Coffee, 15 like both.

Students who like Tea or Coffee = $|A \cup B| = 30 + 25 - 15 = 40$

★ **Exam Tip:** Inclusion-Exclusion numericals (survey-style word problems) are a guaranteed 5-mark question — always draw the Venn diagram alongside your calculation.

Unit I — Important Exam Questions

- 1 Define proposition. Construct the truth table for $(p \rightarrow q) \leftrightarrow (\neg p \vee q)$.
- 2 State and prove De Morgan's Laws for propositions using a truth table.
- 3 Differentiate between converse, inverse and contrapositive with an example.
- 4 What are quantifiers? Explain universal and existential quantifiers with examples. Show how to negate a quantified statement.
- 5 Explain nested quantifiers with an example where order of quantifiers changes the meaning.
- 6 State and explain any five rules of inference with examples.
- 7 Prove by contradiction that $\sqrt{2}$ is an irrational number.
- 8 Prove using direct proof: if n is an even integer, then n^2 is even.
- 9 Define a set. Explain subset, power set and Cartesian product with examples.
- 10 Using a Venn diagram, explain union, intersection, difference and complement of sets.
- 11 State the Principle of Inclusion-Exclusion and solve a numerical using it.
- 12 Prove using set identities: $A - (B \cap C) = (A - B) \cup (A - C)$.

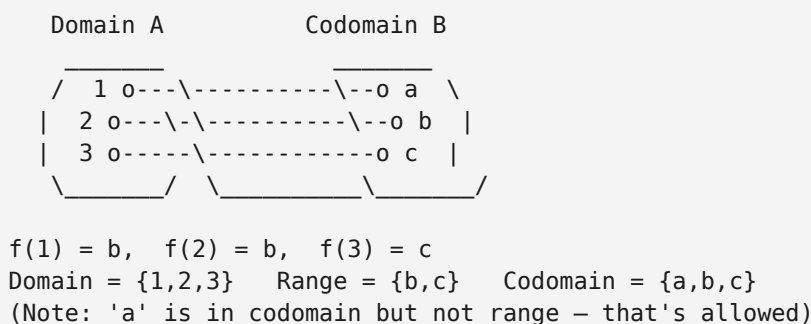
UNIT II

Functions, Sequences & Recurrence Relations

2.1 Functions — Definition & Types

Definition: A **Function** f from set A to set B (written $f: A \rightarrow B$) is a rule that assigns to **each** element of A **exactly one** element of B . A is called the **Domain**, B is the **Codomain**, and the set of actual output values is the **Range** ($\text{Range} \subseteq \text{Codomain}$).

Fig 2.1 — Function Mapping



2.2 One-to-One (Injective) & Onto (Surjective) Functions

One-to-One (Injective) Function

Definition: A function f is **One-to-One (Injective)** if different elements of the domain always map to different elements of the codomain: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. No two domain elements share the same image.

Onto (Surjective) Function

Definition: A function f is **Onto (Surjective)** if every element of the codomain is the image of at least one element of the domain — i.e. $\text{Range} = \text{Codomain}$.

Bijective Function

Definition: A function that is **both One-to-One AND Onto** is called **Bijective (One-to-One Correspondence)** — every element of A maps to a unique element of B and every element of B is covered exactly once.

Fig 2.2 — Injective vs Surjective vs Bijective

One-to-One (not onto)	Onto (not one-to-one)	Bijjective
1 o---o a	1 o---\	1 o---o a
2 o---o b	2 o----o a	2 o---o b
3 o---o c	3 o---/	3 o---o c
o d (unused)	(a is hit twice, no element unused in B)	(perfect pairing, both conditions hold)

Example — Checking Function Types

$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 \rightarrow$ NOT one-to-one ($f(2)=f(-2)=4$), NOT onto (negative numbers have no preimage)

$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 3 \rightarrow$ Both one-to-one AND onto $\rightarrow$ **Bijjective**

2.3 Inverse Functions & Composition of Functions

Inverse Function

Definition: If $f: A \rightarrow B$ is a **bijection**, its **Inverse Function** $f^{-1}: B \rightarrow A$ exists, defined by $f^{-1}(y) = x$ whenever $f(x) = y$. Only bijective functions have a true inverse.

Example — Finding an Inverse

$f(x) = 2x + 3 \rightarrow$ let $y = 2x + 3 \Rightarrow x = (y - 3)/2 \rightarrow f^{-1}(x) = (x - 3)/2$

Composition of Functions

Definition: The **Composition** of $f: A \rightarrow B$ and $g: B \rightarrow C$ is $(g \circ f): A \rightarrow C$ defined by $(g \circ f)(x) = g(f(x))$ — apply f first, then apply g to the result.

Example — Function Composition

$f(x) = x + 1, g(x) = x^2 \rightarrow (g \circ f)(x) = g(f(x)) = g(x + 1) = (x + 1)^2$
 $(f \circ g)(x) = f(g(x)) = f(x^2) = x^2 + 1$ **Note: $g \circ f \neq f \circ g$ in general**

2.4 Partial Functions

Definition: A **Partial Function** f from A to B is a mapping where $f(x)$ is defined for only **some** elements of A , not necessarily all — as opposed to a **Total Function** where every element of the domain has a defined image.

Example — Partial Function

$f(x) = 1/x$ is a partial function on $\mathbb{R}$ because it is **undefined at $x = 0$** — the domain of definition excludes 0.

★ **Exam Tip:** Injective / Surjective / Bijective definitions with small mapping-diagram examples (as in Fig 2.2) are a guaranteed 5-mark question — always draw the arrow diagram.

2.5 Sequences

Definition: A **Sequence** is an ordered list of elements, denoted $\{a_n\}$, where a_n is the n -th term, described either by an explicit formula or a recurrence relation.

Sequence Type	Formula/Rule	Example (first 5 terms)
Arithmetic Progression (AP)	$a_n = a + (n-1)d$	2, 5, 8, 11, 14 ($a=2, d=3$)
Geometric Progression (GP)	$a_n = a \cdot r^{n-1}$	3, 6, 12, 24, 48 ($a=3, r=2$)
Fibonacci Sequence	$a_n = a_{n-1} + a_{n-2}$	0, 1, 1, 2, 3, 5, 8...

2.6 Recurrence Relations

Definition: A **Recurrence Relation** expresses a term of a sequence as a function of one or more of its preceding terms, together with initial condition(s).

Example — Solving a Simple Recurrence

$a_n = a_{n-1} + 3, \quad a_0 = 2$
 $a_1=5, a_2=8, a_3=11 \dots$ General solution: $a_n = 2 + 3n$

Solving Linear Homogeneous Recurrence Relations

For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ (2nd order, constant coefficients), form the **characteristic equation** $x^2 - c_1 x - c_2 = 0$, solve for roots r_1, r_2 , and write the general solution based on the roots.

General Solution Forms

Case 1 (distinct real roots $r_1 \neq r_2$):
 $a_n = A(r_1)^n + B(r_2)^n$

Case 2 (repeated root $r_1 = r_2 = r$):
 $a_n = (A + Bn)(r)^n$

A, B are constants found using the initial conditions.

Example — Solving Fibonacci-style Recurrence

$a_n = a_{n-1} + 2a_{n-2}, \quad a_0=1, a_1=1$
Characteristic equation: $x^2 - x - 2 = 0 \rightarrow (x-2)(x+1)=0 \rightarrow$ roots $r_1=2, r_2=-1$
General solution: $a_n = A(2)^n + B(-1)^n$ — solve A, B using $a_0=1, a_1=1$

2.7 Summations

Definition: A **Summation** Σ represents the sum of a sequence of terms following a given pattern, over a specified range of the index.

Summation Notation

Sigma notation: $\sum (i = 1 \text{ to } n) a_i = a_1 + a_2 + \dots + a_n$

Standard Summation Formulas

Series	Formula
$\sum (i=1 \text{ to } n) 1$	$= n$
$\sum (i=1 \text{ to } n) i$	$= n(n+1)/2$
$\sum (i=1 \text{ to } n) i^2$	$= n(n+1)(2n+1)/6$
$\sum (i=1 \text{ to } n) i^3$	$= [n(n+1)/2]^2$
$\sum (i=0 \text{ to } n) r^i (r \neq 1)$	$= (r^{n+1} - 1)/(r - 1)$ (Geometric Series)

Example — Summation Numerical

Find $\sum (i=1 \text{ to } 5) i^2 = 1+4+9+16+25 = \mathbf{55}$ (Check with formula: $5 \cdot 6 \cdot 11 / 6 = 330 / 6 = 55 \checkmark$)

★ **Exam Tip:** Memorise the four standard summation formulas above ($\sum i$, $\sum i^2$, $\sum i^3$, geometric series sum) — they appear repeatedly in counting and complexity analysis problems across the paper.

Unit II — Important Exam Questions

- 1 Define function. Explain domain, codomain and range with an example.
- 2 Differentiate between one-to-one, onto and bijective functions with suitable examples/diagrams.
- 3 What is an inverse function? Find the inverse of $f(x) = 3x - 5$.
- 4 Explain composition of functions with an example. Show that $(g \circ f) \neq (f \circ g)$ in general.
- 5 What is a partial function? Differentiate it from a total function.
- 6 Define a sequence. Differentiate between arithmetic and geometric progressions with examples.
- 7 What is a recurrence relation? Solve $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0=1$, $a_1=4$.
- 8 State and prove the formula for the sum of the first n natural numbers.
- 9 Evaluate $\sum (i=1 \text{ to } 10) i$ and $\sum (i=1 \text{ to } 10) i^2$ using standard formulas.

UNIT III Counting Techniques & Relations

3.1 Basics of Counting

Sum Rule & Product Rule

Rule	Statement	Example
Sum Rule	If a task can be done in m ways OR n ways (no overlap), total ways = $m + n$	Choose a fruit: 3 apples or 4 mangoes → $3+4=7$ ways
Product Rule	If a task consists of a sequence of k independent sub-tasks with $n_1, n_2, \dots, n_k$ ways each, total ways = $n_1 \times n_2 \times \dots \times n_k$	3 shirts $\times$ 2 pants = 6 outfit combinations

Example — Sum vs Product Rule

A password is 1 letter (26 choices) followed by 1 digit (10 choices) → Product Rule: $26 \times 10 = 260$ possible passwords.

3.2 The Pigeonhole Principle

Definition: The **Pigeonhole Principle** states: if n pigeons are placed into k pigeonholes and $n > k$, then at least one pigeonhole must contain **more than one** pigeon.

Fig 3.1 — Pigeonhole Principle

Pigeonholes: [1] [2] [3] <- only 3 holes
Pigeons : * * * * <- 4 pigeons to place

By Pigeonhole Principle, at least one hole must get ≥ 2 pigeons
(since 4 pigeons $>$ 3 holes)

Generalized Pigeonhole Principle

Generalized Pigeonhole Formula

If n pigeons are placed into k pigeonholes,
at least one pigeonhole contains at least $\lceil n / k \rceil$ pigeons
(ceiling of n divided by k)

Example — Pigeonhole Numerical

Among 100 people, show at least $\lceil 100/12 \rceil = 9$ were born in the same month.
100 people, 12 months (pigeonholes) → $\lceil 100/12 \rceil = \lceil 8.33 \rceil = 9$ people share a birth month at minimum.

★ **Exam Tip:** Pigeonhole Principle 'prove that...' word problems are extremely common — always identify what the 'pigeons' and 'pigeonholes' are first, then apply the ceiling formula.

3.3 Permutations & Combinations

Permutations (Order Matters)

Definition: A **Permutation** is an ordered arrangement of objects. The number of permutations of n distinct objects taken r at a time is denoted $P(n,r)$ or ${}^n P_r$.

Permutation Formula

$$P(n, r) = n! / (n - r)!$$

$$\text{When } r = n: P(n, n) = n!$$

Combinations (Order Does NOT Matter)

Definition: A **Combination** is a selection of objects where order does not matter, denoted $C(n,r)$ or ${}^n C_r$ or $(n \text{ choose } r)$.

Combination Formula

$$C(n, r) = n! / [r! (n - r)!] = P(n,r) / r!$$

Example — Permutation vs Combination

From 5 people, choose a President and a Secretary (order matters, 2 different roles): $P(5,2) = 5!/3! = \mathbf{20}$ ways.

From 5 people, choose a committee of 2 (order doesn't matter): $C(5,2) = 5!/(2!3!) = \mathbf{10}$ ways.

Permutations with Repetition

Permutation with Repetition Formula

Number of arrangements of n objects where there are n_1 objects of type 1, n_2 of type 2, ..., n_k of type k :

$$n! / (n_1! \times n_2! \times \dots \times n_k!)$$

Example — Permutation with Repetition

Arrangements of the letters in 'MISSISSIPPI' (11 letters: M-1, I-4, S-4, P-2)
 $= 11! / (1! 4! 4! 2!) = 39916800 / 1152 = \mathbf{34650}$

3.4 Binomial Coefficients & Identities

Definition: The **Binomial Theorem** expands $(x + y)^n$ using binomial coefficients $C(n,r)$, which count the ways to choose r items from n .

Binomial Theorem

$$(x + y)^n = \sum_{r=0}^n C(n, r) x^{n-r} y^r$$

Example — Binomial Expansion

$$\begin{aligned}(x + y)^4 &= C(4,0)x^4 + C(4,1)x^3y + C(4,2)x^2y^2 + C(4,3)xy^3 + C(4,4)y^4 \\ &= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4\end{aligned}$$

Important Binomial Identities

Identity	Statement
Pascal's Identity	$C(n,r) = C(n-1,r-1) + C(n-1,r)$
Symmetry	$C(n,r) = C(n, n-r)$
Sum of Row	$\sum_{r=0}^n C(n,r) = 2^n$
Vandermonde's Identity	$C(m+n, r) = \sum_{k=0}^r C(m,k) \cdot C(n,r-k)$

Fig 3.2 — Pascal's Triangle

Pascal's Triangle (rows = n, each entry = $C(n,r)$)

```

n=0:      1
n=1:     1 1
n=2:    1 2 1
n=3:   1 3 3 1
n=4:  1 4 6 4 1

```

Each number = sum of the two numbers diagonally above it
(Pascal's Identity in visual form)

★ **Exam Tip:** Permutation vs Combination confusion is the #1 mistake students make — ask yourself 'does order matter?' If YES → Permutation. If NO → Combination.

3.5 Relations & their Properties

Definition: A **Binary Relation** R from set A to set B is a subset of $A \times B$. If $(a,b) \in R$, we write $a R b$. When $A = B$, R is called a relation **on A**.

Properties of Relations (on a set A)

Property	Definition	Example (A = integers)
Reflexive	$\forall a \in A, (a,a) \in R$	' $\leq$ ' is reflexive: $a \leq a$ always
Symmetric	$(a,b) \in R \Rightarrow (b,a) \in R$	'is sibling of' is symmetric
Antisymmetric	$(a,b) \in R \text{ and } (b,a) \in R \Rightarrow a=b$	' $\leq$ ' is antisymmetric
Transitive	$(a,b) \in R \text{ and } (b,c) \in R \Rightarrow (a,c) \in R$	' $<$ ' is transitive

Example — Checking Relation Properties

$R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$ on $A = \{1, 2, 3\}$

Reflexive? Yes (all (a,a) present). Symmetric? Yes $((1,2)$ and $(2,1)$ both present). Transitive? Yes. → This is an **Equivalence Relation**.

3.6 n-ary Relations & their Applications

Definition: An **n-ary Relation** on sets $A_1, A_2, \dots, A_n$ is a subset of $A_1 \times A_2 \times \dots \times A_n$. A **Binary Relation** is the special case $n=2$ (most common). n-ary relations are the theoretical foundation of relational database tables.

Example — n-ary Relation Application

A database table STUDENT(RollNo, Name, Dept) is a **ternary (3-ary) relation** — each row is a tuple in $\text{RollNo} \times \text{Name} \times \text{Dept}$.

3.7 Representing Relations & Closure of Relations

Representing Relations

Method	Description
Set of Ordered Pairs	$R = \{(1,2), (2,3), (1,3)\}$
Matrix Representation	Boolean/0-1 matrix M where $M[i][j]=1$ if $(i,j) \in R$
Directed Graph (Digraph)	Vertices = elements, directed edge for each pair in R

Fig 3.3 — Matrix and Digraph Representation of a Relation

$R = \{(1,2), (2,3), (1,3)\}$ on $A = \{1, 2, 3\}$

Matrix:

```

  1  2  3
1[ 0  1  1 ]
2[ 0  0  1 ]
3[ 0  0  0 ]

```

Digraph:

```

  1 -----> 2
    \       |
     \       v
      ---> 3

```

Closure of Relations

Definition: The **Closure** of a relation R with respect to a property (reflexive, symmetric, transitive) is the smallest relation containing R that has that property.

Closure Type	How to Compute
Reflexive Closure	Add all missing pairs (a,a) to R
Symmetric Closure	Add (b,a) for every $(a,b) \in R$
Transitive Closure	Keep adding (a,c) whenever $(a,b), (b,c) \in R$ until no more changes (Warshall's Algorithm is commonly used to compute this)

Example — Transitive Closure Example

$R = \{(1,2), (2,3)\} \rightarrow$ since $(1,2)$ and $(2,3) \in R$, add $(1,3)$.
 Transitive Closure = $\{(1,2), (2,3), (1,3)\}$

3.8 Equivalence Relations

Definition: A relation that is **Reflexive, Symmetric AND Transitive** is called an **Equivalence Relation**. It partitions the set into disjoint groups called **Equivalence Classes**.

Example — Equivalence Relation Example

$R: 'a \equiv b \pmod{3}'$ on integers (a and b leave the same remainder when divided by 3).
 Equivalence classes: $[0] = \{\dots, -3, 0, 3, 6, \dots\}$, $[1] = \{\dots, -2, 1, 4, 7, \dots\}$, $[2] = \{\dots, -1, 2, 5, 8, \dots\}$ — every integer belongs to exactly one class.

★ **Exam Tip:** To prove a relation is an equivalence relation, you **MUST** verify all three properties (reflexive, symmetric, transitive) separately — missing even one means it is **NOT** an equivalence relation.

3.9 Partial Orderings

Definition: A relation that is **Reflexive, Antisymmetric AND Transitive** is called a **Partial Order**, and the set with this relation is a **Partially Ordered Set (POSET)**, written $(A, \leq)$.

Example — Partial Order Example

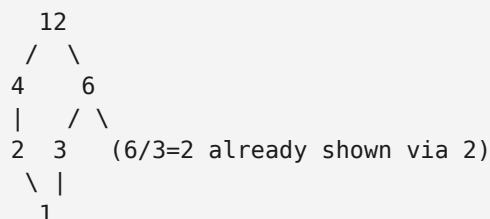
The 'divides' relation ($a \mid b$) on positive integers is a partial order: reflexive ($a \mid a$), antisymmetric ($a \mid b$ and $b \mid a \Rightarrow a = b$), transitive ($a \mid b$ and $b \mid c \Rightarrow a \mid c$).

Hasse Diagram

Definition: A **Hasse Diagram** is a simplified visual representation of a partial order — it omits reflexive loops and transitive edges, showing only the direct 'covers' relationships, with 'greater' elements drawn higher.

Fig 3.4 — Hasse Diagram for Divisibility on $\{1, 2, 3, 4, 6, 12\}$

Divisibility poset on $\{1, 2, 3, 4, 6, 12\}$:



Read bottom-up: 1 divides everything, 12 is divided by everything below it connected by an ascending path.

★ **Exam Tip:** Equivalence Relation vs Partial Order is a favourite 'differentiate' question: both need Reflexive + Transitive, but Equivalence needs SYMMETRIC while Partial Order needs ANTISYMMETRIC instead.

Unit III — Important Exam Questions

- 1 State and explain the sum rule and product rule of counting with examples.
- 2 State the Pigeonhole Principle. Prove that among any 13 people, at least two share a birth month.
- 3 Differentiate between permutation and combination with examples. In how many ways can a committee of 3 be chosen from 8 people?
- 4 State the Binomial Theorem and expand $(x+y)^5$.
- 5 Derive Pascal's Identity and illustrate it using Pascal's Triangle.
- 6 Define a relation. Explain reflexive, symmetric, antisymmetric and transitive properties with examples.
- 7 What is the closure of a relation? Find the transitive closure of $R=\{(1,2),(2,3),(3,4)\}$ on $\{1,2,3,4\}$.
- 8 What is an equivalence relation? Show that 'congruence modulo n ' is an equivalence relation and find its equivalence classes for $n=3$.
- 9 Define a partial order and a POSET. Draw the Hasse diagram for the divisibility relation on $\{1,2,3,4,6,12\}$.
- 10 Represent a given relation using a matrix and a directed graph.

UNIT IV Graph Theory & Trees

4.1 Graph Terminology & Special Types of Graphs

Definition: A **Graph** $G = (V, E)$ consists of a set of **Vertices (V)** and a set of **Edges (E)**, where each edge connects a pair of vertices.

Fig 4.1 — Simple Undirected Graph



$V = \{1, 2, 3, 4\}$ $E = \{(1, 2), (1, 3), (1, 4), (2, 3), (3, 4)\}$
 $\deg(1) = 3, \deg(2) = 2, \deg(3) = 3, \deg(4) = 2$

Basic Terminology

Term	Meaning
Degree of a vertex	Number of edges incident to (touching) that vertex
Adjacent Vertices	Two vertices connected directly by an edge
Path	A sequence of edges connecting a sequence of distinct vertices
Cycle	A path that starts and ends at the same vertex, no repeated edges
Connected Graph	There exists a path between every pair of vertices
Simple Graph	No loops (self-edges) and no multiple/parallel edges
Multigraph	Multiple edges allowed between the same pair of vertices

Handshaking Theorem

Handshaking Theorem

Sum of degrees of all vertices = $2 \times (\text{number of edges})$

$$\sum \deg(v) = 2|E|$$

Consequence: the number of vertices with ODD degree is always EVEN.

Example — Handshaking Theorem Check

Fig 4.1: $\deg \text{ sum} = 3+2+3+2 = 10 = 2 \times 5 \text{ edges} \checkmark$ Odd-degree vertices: 1 and 3 (two of them — even count) $\checkmark$

Special Types of Graphs

Graph Type	Description
Complete Graph K_n	Every pair of the n vertices is connected; has $n(n-1)/2$ edges
Cycle Graph C_n	n vertices connected in a single closed loop
Wheel Graph W_n	A cycle C_n plus one central vertex connected to all others
Regular Graph	Every vertex has the same degree
Null/Empty Graph	A graph with vertices but no edges

Fig 4.2 — K_4 , C_4 and W_4 (Special Graphs) K_4 (Complete Graph)

```

A --- B
|\  /|
| X |
|/  \|
C --- D

```

 C_4 (Cycle Graph)

```

A --- B
|     |
|     |
|     |
D --- C

```

 W_4 (Wheel Graph)

```

A --- B
|\  /|
| C |
|/  \|
D --- E

```

(C = hub, connected to all)

4.2 Bipartite Graphs & Representing Graphs

Definition: A graph is **Bipartite** if its vertices can be split into **two disjoint sets** V_1 and V_2 such that every edge connects a vertex in V_1 to a vertex in V_2 (no edges within the same set). A **Complete Bipartite Graph** $K_{m,n}$ connects every vertex of V_1 (size m) to every vertex of V_2 (size n).

Fig 4.3 — Bipartite Graph

```

V1      V2
A  ----- X
B X ----- Y
C  ----- Z

```

Test for Bipartite:
A graph is bipartite IFF it
contains NO ODD-LENGTH CYCLE.

($K_{3,3}$ type bipartite graph – no edges within V_1 or within V_2)

Representing Graphs (Recap)

Representation	Structure
Adjacency Matrix	$n \times n$ matrix; entry $(i,j)=1$ if edge exists between i,j , else 0
Adjacency List	Each vertex stores a list of its directly connected neighbours

4.3 Isomorphism of Graphs

Definition: Two graphs G_1 and G_2 are **Isomorphic** if there exists a one-to-one correspondence (bijection) between their vertex sets that **preserves adjacency** — essentially, they are the 'same graph' drawn differently.

Necessary Conditions for Isomorphism (all must hold)

- Same number of vertices.
- Same number of edges.
- Same degree sequence (sorted list of vertex degrees must match).
- Same number of connected components.
- Corresponding cycles of the same length must exist in both graphs.

Example — Isomorphism Check

G1 degree sequence: {2,2,2,2} (a 4-cycle) G2 degree sequence: {1,1,2,2} (a path) → **NOT isomorphic** (degree sequences differ).

★ **Exam Tip:** Isomorphism 'necessary but not sufficient' conditions are commonly tested — matching vertex/edge count and degree sequence does NOT guarantee isomorphism, but MISMATCHING them proves graphs are NOT isomorphic.

4.4 Euler & Hamilton Paths

Euler Path & Circuit

Definition: An **Euler Path** traverses **every edge exactly once** (vertices may repeat). An **Euler Circuit** is an Euler Path that starts and ends at the same vertex.

Euler's Theorem (Condition for Euler Path/Circuit)

A connected graph has an EULER CIRCUIT $\Leftrightarrow$ every vertex has EVEN degree.
 A connected graph has an EULER PATH (not circuit)
 $\Leftrightarrow$ exactly TWO vertices have ODD degree.

Example — Euler Path — Königsberg Bridge Problem

The famous 7-bridges-of-Königsberg problem (Euler, 1736) has 4 landmasses all with ODD degree → **no Euler path exists** — this founded graph theory as a subject.

Hamilton Path & Circuit

Definition: A **Hamilton Path** visits **every vertex exactly once** (edges may be skipped). A **Hamilton Circuit** is a Hamilton Path that returns to the starting vertex. Unlike Euler paths, there is **no simple necessary-and-sufficient condition** for Hamilton paths — it is generally checked case by case.

Basis	Euler Path/Circuit	Hamilton Path/Circuit
Covers	Every EDGE exactly once	Every VERTEX exactly once
Existence Test	Simple degree-based theorem	No simple general test (NP-Complete)
Vertices/Edges may repeat?	Vertices may repeat	Edges may be skipped

★ **Exam Tip:** Euler vs Hamilton is a guaranteed 'differentiate' question — remember: Euler = every EDGE once; Hamilton = every VERTEX once. Euler has an easy degree-based test; Hamilton does not.

4.5 Shortest Path Problem — Dijkstra's Algorithm

Definition: Dijkstra's Algorithm finds the **shortest path** from a single source vertex to all other vertices in a **weighted graph with non-negative edge weights**, using a greedy approach.

- 1 Assign distance 0 to the source vertex, ∞ to all others; mark all vertices unvisited.
- 2 Pick the unvisited vertex with the smallest known distance.
- 3 For each unvisited neighbour, calculate distance via current vertex; if smaller than the neighbour's known distance, update it.
- 4 Mark the current vertex as visited (its shortest distance is now finalised).
- 5 Repeat steps 2-4 until all vertices are visited.

Fig 4.4 — Dijkstra's Algorithm Trace

Graph:

A	--2--	B
4		1
C	--3--	D

Dijkstra from A:

dist[A]=0

Visit A: update B=2, C=4

Visit B (dist 2, smallest): update D = 2+1 = 3

Visit D (dist 3): update C = min(4, 3+3=6) = 4 (no change)

Visit C (dist 4): done

Shortest distances from A: A=0, B=2, D=3, C=4

★ **Exam Tip:** Dijkstra's Algorithm trace questions (given a weighted graph, find shortest path from a source) are near-guaranteed — practice maintaining a distance-table and updating it step by step as shown above.

4.6 Traveling Salesperson Problem (TSP)

Definition: The **Traveling Salesperson Problem** asks: given a set of cities and distances between every pair, find the **shortest possible route** that visits every city exactly once and returns to the starting city — essentially finding the minimum-weight **Hamilton Circuit**.

- TSP is an **NP-Hard** problem — no known polynomial-time algorithm solves it exactly for all cases.
- Brute force checks all $(n-1)!/2$ possible routes for n cities — becomes impractical very quickly as n grows.
- Approximation approaches: Nearest Neighbour heuristic, Minimum Spanning Tree-based approximation.

4.7 Planar Graphs & Graph Coloring

Planar Graphs

Definition: A graph is **Planar** if it can be drawn on a plane such that **no two edges cross** each other (except at shared vertices).

Euler's Formula for Planar Graphs

Euler's Formula for connected planar graphs:

$$V - E + F = 2$$

where V = vertices, E = edges, F = faces (including the outer/unbounded face)

Example — Euler's Formula Check

Cube graph: $V=8$, $E=12$, $F=6$ (faces of the cube) + 1 outer face = 7? Let's recompute: actually for the cube graph $V=8$, $E=12$, $F=6 \rightarrow 8-12+6 = 2$ ✓ formula holds.

Graph Coloring

Definition: Graph Coloring assigns colors to vertices such that **no two adjacent vertices share the same color**. The **Chromatic Number** $\chi(G)$ is the minimum number of colors needed.

Graph Type	Chromatic Number
Bipartite Graph	2 (always, if it has at least one edge)
Complete Graph K_n	n
Cycle Graph C_n (n even)	2
Cycle Graph C_n (n odd)	3
Any Planar Graph	≤ 4 (Four Color Theorem)

★ **Exam Tip:** The **Four Color Theorem** — every planar graph (e.g. a map) can be colored with at most 4 colors so no two adjacent regions share a color — is a famous theoretical result worth remembering as a 'write short note' answer.

4.8 Tree Traversal & Minimum Spanning Trees

Trees (Recap for Graph Theory Context)

Definition: A **Tree** is a **connected, acyclic** graph. A tree with n vertices always has exactly $n - 1$ edges. A **Forest** is a disjoint union of trees (acyclic but possibly disconnected).

Tree Traversal

Tree traversal (Inorder, Preorder, Postorder for binary trees; also Level-order/BFS) visits every node exactly once — see the Data Structures notes Unit IV for the full recursive algorithms and dry-run examples on a sample tree.

Spanning Tree & Minimum Spanning Tree (MST)

Definition: A **Spanning Tree** of a connected graph G is a subgraph that includes **all vertices** of G and is a **tree** (connected, no cycles, uses exactly $V-1$ edges). A **Minimum Spanning Tree** is the spanning tree with the **least total edge weight** among all possible spanning trees.

Kruskal's Algorithm (Greedy, Edge-based)

- 1 Sort all edges in increasing order of weight.
- 2 Pick the smallest edge; if it does NOT form a cycle with the already-chosen edges, add it to the MST (cycle check done using Union-Find/Disjoint Set).
- 3 Repeat until the MST has exactly $(V - 1)$ edges.

Prim's Algorithm (Greedy, Vertex-based)

- 1 Start from any vertex; mark it as part of the MST.
- 2 Among all edges connecting an MST vertex to a non-MST vertex, pick the minimum-weight edge and add both the edge and new vertex to the MST.
- 3 Repeat until all vertices are included in the MST.

Fig 4.5 — Kruskal's Algorithm Trace (Minimum Spanning Tree)

Graph (weighted):

```

      A --1-- B
      | \    |
      4 2    3
      |  \   |
      C --5-- D
  
```

Kruskal's trace (sort edges: $AB=1, AC=2, BD=3, AD=4, CD=5$):
 Add $AB(1)$: no cycle. Add $AC(2)$: no cycle.
 Add $BD(3)$: no cycle. MST now has 3 edges = $V-1$ ($V=4$) -> STOP
 MST edges = $\{AB, AC, BD\}$, Total weight = $1+2+3 = 6$

★ **Exam Tip:** Kruskal's vs Prim's is a common 'differentiate' question: Kruskal's picks the globally smallest edge each time (edge-focused, needs cycle detection); Prim's grows one connected tree outward from a start vertex (vertex-focused). Both are Greedy algorithms and both give a correct MST.

Unit IV — Important Exam Questions

- 1 Define a graph. Explain the Handshaking Theorem with an example.
- 2 Explain special types of graphs: complete graph, cycle graph, wheel graph and regular graph with diagrams.
- 3 What is a bipartite graph? Explain with an example and state the test for bipartiteness using odd cycles.
- 4 What is graph isomorphism? State the necessary conditions for two graphs to be isomorphic.
- 5 Differentiate between Euler path/circuit and Hamilton path/circuit with examples.

- 6 State Euler's theorem for the existence of an Euler circuit. Explain using the Königsberg bridge problem.
- 7 Explain Dijkstra's algorithm with a suitable example and find the shortest path from a given source vertex.
- 8 What is the Traveling Salesperson Problem? Why is it considered computationally hard?
- 9 Define a planar graph. State and verify Euler's formula ($V-E+F=2$) for a given graph.
- 10 What is graph coloring? Define chromatic number and state the Four Color Theorem.
- 11 Define a spanning tree and minimum spanning tree. Find the MST of a given weighted graph using Kruskal's algorithm.
- 12 Differentiate between Kruskal's and Prim's algorithms for finding an MST.

QUICK REVISION Formula & Identity Cheat-Sheet

Logic & Sets — Quick Formulas

Concept	Formula / Rule
De Morgan (Logic)	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
De Morgan (Quantifiers)	$\neg \forall x P(x) \equiv \exists x \neg P(x)$ $\neg \exists x P(x) \equiv \forall x \neg P(x)$
Contrapositive	$p \rightarrow q \equiv \neg q \rightarrow \neg p$
Power Set Size	$ P(A) = 2^n$, where $n = A $
Cartesian Product Size	$ A \times B = A \times B $
Inclusion-Exclusion (2 sets)	$ A \cup B = A + B - A \cap B $

Counting & Relations — Quick Formulas

Concept	Formula / Rule
Permutation	$P(n, r) = n! / (n-r)!$
Combination	$C(n, r) = n! / [r!(n-r)!]$
Pigeonhole (generalized)	At least $\lceil n/k \rceil$ pigeons in one hole (n pigeons, k holes)
Sum of first n numbers	$\sum i = n(n+1)/2$
Sum of first n squares	$\sum i^2 = n(n+1)(2n+1)/6$
Equivalence Relation	Reflexive + Symmetric + Transitive
Partial Order	Reflexive + Antisymmetric + Transitive

Graph Theory — Quick Formulas

Concept	Formula / Rule
Handshaking Theorem	$\sum \deg(v) = 2 E $
Complete Graph Edges	K_n has $n(n-1)/2$ edges
Euler Circuit exists iff	Every vertex has EVEN degree
Euler Path (not circuit) exists iff	Exactly TWO vertices have ODD degree
Tree Edge Count	A tree with n vertices has exactly $(n-1)$ edges
Euler's Planar Formula	$V - E + F = 2$
Bipartite Test	Graph is bipartite iff it has NO odd-length cycle

One-Line Definitions to Remember

- **Tautology:** Always True. **Contradiction:** Always False.
- **Injective:** No two inputs share an output. **Surjective:** Every output is used. **Bijjective:** Both.
- **Permutation:** Order matters. **Combination:** Order doesn't matter.
- **Equivalence Relation:** Reflexive + Symmetric + Transitive.
- **Partial Order:** Reflexive + Antisymmetric + Transitive.
- **Euler Path/Circuit:** Uses every EDGE once. **Hamilton Path/Circuit:** Visits every VERTEX once.
- **Spanning Tree:** Connects all vertices, no cycles, exactly $V-1$ edges.
- **Chromatic Number:** Minimum colors so adjacent vertices differ.

Last-Minute Exam Strategy

- 1 For every definition-based question, write the exact definition first, then one line of explanation, then a diagram/truth table wherever possible.
- 2 For 'prove/derive' questions (Pigeonhole, $\sqrt{2}$ irrational, De Morgan), write the proof in clear numbered steps — partial marks are given for each correct step.
- 3 For counting word problems, first decide Permutation vs Combination vs Pigeonhole before writing any formula.
- 4 For graph theory numericals, always draw the graph first, then label vertex degrees before answering Euler/Hamilton/Dijkstra/MST questions.
- 5 Revise the Formula Cheat-Sheet on this page one final time before entering the exam hall.

End of Notes — Digital Notes Prepared by Manthan Sahu, Rajendra University, Balangir. All the best for your examination!

DMS — EXAM BOOSTER ADDENDUM

Discrete Mathematical Structures • Semester III • Paper V

Purpose: strengthen the original notes against the official syllabus by adding more solved numericals, a standalone Tree Traversal section, Graph Coloring practice, and 1/2/5/8-mark exam preparation.

Added area	What is added
Unit II	Extra recurrence-relation numericals with complete steps
Unit III	Extra Pigeonhole, P&C; and relation-closure practice
Unit IV	Standalone Tree Traversal + Graph Coloring + Dijkstra practice
All Units	1-mark, 2-mark, 5-mark and 8-mark question bank with answer points

Important: This is an addendum to the original notes, not a replacement. The original notes already cover the official four-unit structure and include quick revision material and unit-wise important questions.

UNIT II — Recurrence Relations: Extra Practice

The original notes already introduce first-order recurrences and second-order homogeneous recurrences using characteristic roots. This addendum adds more step-by-step exam numericals.

1. First-order recurrence

Solve: $a_n = a_{n-1} + 4$, $a_0 = 3$.

1. Expand: $a_1=7$, $a_2=11$, $a_3=15$.

2. Observe that 4 is added at every step.

3. Hence $a_n = 3 + 4n$.

Answer: $a_n = 4n + 3$.

2. Second-order recurrence with distinct roots

Solve: $a_n = 5a_{n-1} - 6a_{n-2}$, $a_0=1$, $a_1=4$.

1. Characteristic equation: $r^2 - 5r + 6 = 0$.

2. Factor: $(r-2)(r-3)=0$, so $r=2,3$.

3. General solution: $a_n = A2^n + B3^n$.

4. Use $a_0=1$: $A+B=1$.

5. Use $a_1=4$: $2A+3B=4$.

6. Subtract $2(A+B)=2$ from the second equation: $B=2$, so $A=-1$.

Answer: $a_n = -2^n + 2 \cdot 3^n$.

3. Repeated-root case

Solve: $a_n = 6a_{n-1} - 9a_{n-2}$, $a_0=2$, $a_1=3$.

1. Characteristic equation: $r^2 - 6r + 9 = 0 = (r-3)^2$.

2. Repeated root $r=3$, so $a_n = (A+Bn)3^n$.

3. $n=0$ gives $A=2$.

4. $n=1$ gives $(2+B)3=3$, so $B=-1$.

Answer: $a_n = (2-n)3^n$.

UNIT II — More Numericals & Quick Methods

4. Summation practice

Evaluate $\sum_{i=1}^{10} i$ and $\sum_{i=1}^{10} i^2$.

1. $\sum i = n(n+1)/2 = 10 \times 11/2 = 55$.

2. $\sum i^2 = n(n+1)(2n+1)/6 = 10 \times 11 \times 21/6 = 385$.

Answer: 55 and 385.

5. Function-type identification

Function	Result	Reason
$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$	Neither injective nor surjective	$f(2) = f(-2)$; negative outputs are not reached
$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 3$	Bijjective	Distinct inputs give distinct outputs and every real output has a preimage
$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$	Bijjective	Strictly increasing and every real value occurs

6. Inverse-function numerical

Find the inverse of $f(x) = 3x - 5$.

1. Let $y = 3x - 5$.

2. $y + 5 = 3x$, hence $x = (y + 5)/3$.

3. Replace y by x .

Answer: $f^{-1}(x) = (x + 5)/3$.

7. Exam checklist for Unit II

- For function questions, always identify domain, codomain and range.
- For injective/onto questions, use either algebra or a clear mapping diagram.
- For recurrence numericals, write the characteristic equation before writing the general solution.
- For repeated roots, use $(A + Bn)r^n$.
- For summations, first identify which standard formula applies.

UNIT III — Counting: Extra Solved Numericals

1. Pigeonhole Principle

Show that among 25 students, at least 3 have birthdays in the same month.

1. There are 12 months (pigeonholes) and 25 students (pigeons).
2. Generalized principle gives at least $\text{ceil}(25/12) = \text{ceil}(2.083...) = 3$.

Answer: At least 3 students share a birth month.

2. Permutation

In how many ways can 4 different books be arranged from 7 different books?

Order matters, so use $P(7,4) = 7!/(7-4)! = 7!/3! = 7 \times 6 \times 5 \times 4 = \mathbf{840}$.

3. Combination

How many committees of 3 can be selected from 8 students?

Order does not matter, so $C(8,3) = 8!/(3!5!) = \mathbf{56}$.

4. Password counting

A password contains 2 uppercase letters followed by 2 digits. Repetition is allowed. How many passwords?

$26 \times 26 \times 10 \times 10 = \mathbf{67,600}$.

5. Binomial coefficient

Find the coefficient of x^3y^2 in $(x+y)^5$.

General term: $C(5,r)x^{5-r}y^r$. For x^3y^2 , $r=2$. Coefficient = $C(5,2) = \mathbf{10}$.

Exam rule: **order matters** → **permutation**; **order does not matter** → **combination**.

UNIT III — Relations: Closure, Equivalence & POSET Practice

1. Test relation properties

Let $A=\{1,2,3\}$ and $R=\{(1,1),(2,2),(3,3),(1,2),(2,1)\}$.

Property	Check	Result
Reflexive	$(1,1),(2,2),(3,3)$ are present	Yes
Symmetric	$(1,2)$ and $(2,1)$ both present	Yes
Antisymmetric	$(1,2)$ and $(2,1)$ are both present but $1 \neq 2$	No
Transitive	All required chains remain in R	Yes

Conclusion: R is an equivalence relation because it is reflexive, symmetric and transitive.

2. Transitive closure

Let $R=\{(1,2),(2,3),(3,4)\}$ on $A=\{1,2,3,4\}$.

1. From $(1,2)$ and $(2,3)$, add $(1,3)$.
2. From $(2,3)$ and $(3,4)$, add $(2,4)$.
3. From $(1,3)$ and $(3,4)$, add $(1,4)$.

Transitive closure: $\{(1,2),(2,3),(3,4),(1,3),(2,4),(1,4)\}$.

3. Equivalence relation vs partial order

Concept	Required properties	Key difference
Equivalence relation	Reflexive + Symmetric + Transitive	Uses symmetry; creates equivalence classes
Partial order	Reflexive + Antisymmetric + Transitive	Uses antisymmetry; forms a POSET/Hasse diagram

4. Hasse-diagram exam method

- Start with the relation and list comparable pairs.
- Remove reflexive loops.
- Remove edges implied by transitivity.
- Place smaller/lower elements below larger/higher elements.

UNIT IV — Tree Traversal: Standalone Exam Section

The original notes mention Inorder, Preorder and Postorder, but refer to DSA notes for full algorithms/dry-runs. This addendum makes Tree Traversal standalone for DMS revision.

Sample binary tree

```
A
 / \
B   C
/\  /
D E F
```

Traversal	Rule	For the sample tree
Preorder	Root → Left → Right	A, B, D, E, C, F
Inorder	Left → Root → Right	D, B, E, A, F, C
Postorder	Left → Right → Root	D, E, B, F, C, A
Level-order	Level by level (BFS)	A, B, C, D, E, F

Recursive algorithm patterns

Preorder: visit root; traverse left subtree; traverse right subtree.

Inorder: traverse left subtree; visit root; traverse right subtree.

Postorder: traverse left subtree; traverse right subtree; visit root.

Important exam point

- Each traversal visits every node exactly once.
- For a Binary Search Tree, inorder traversal gives the keys in sorted order.
- Do not confuse traversal order with graph traversal terminology: level-order corresponds to BFS-style traversal.

UNIT IV — Graph Coloring & Dijkstra Practice

1. Graph coloring

A proper vertex coloring assigns colors so that adjacent vertices receive different colors. The chromatic number $\chi(G)$ is the minimum number of colors required.

Graph	Chromatic number	Reason
Non-empty bipartite graph	2	Two parts can receive two different colors
Complete graph K_n	n	Every pair of vertices is adjacent
Even cycle C_n	2	Alternating two colors works
Odd cycle C_n	3	Two-color alternation fails at the closing edge

2. Coloring example

For C_5 (a 5-cycle), try alternating colors 1,2,1,2. The final vertex is adjacent to both ends, so a third color is necessary.

Answer: $\chi(C_5)=3$.

3. Dijkstra numerical

Edges: $A-B=4$, $A-C=1$, $C-B=2$, $B-D=1$, $C-D=5$. Find shortest distances from A.

Step	Chosen vertex	Best known distances (A,B,C,D)
Start	—	0, ∞ , ∞ , ∞
1	A	0, 4, 1, ∞
2	C	0, 3, 1, 6
3	B	0, 3, 1, 4
4	D	0, 3, 1, 4

Shortest paths: $A \rightarrow C = 1$, $A \rightarrow C \rightarrow B = 3$, $A \rightarrow C \rightarrow B \rightarrow D = 4$.

Reminder: Dijkstra's algorithm assumes non-negative edge weights.

UNIT IV — MST, Planar Graphs & Euler/Hamilton Revision

Kruskal vs Prim

Feature	Kruskal	Prim
Focus	Edges	Vertices / growing tree
Starting point	No fixed start vertex required	Starts from a chosen vertex
Main choice	Globally smallest safe edge	Smallest edge leaving current MST
Cycle handling	Explicit cycle check	Growth naturally avoids adding an internal cycle

MST numerical reminder

For a connected graph with V vertices, every spanning tree has exactly $V-1$ edges. For $V=6$, an MST must contain **5 edges**.

Planar graph check

For a connected planar graph, $V-E+F=2$. If $V=8$ and $E=12$, then $F=2-8+12=6$.

Euler vs Hamilton

Euler	Hamilton
Uses every edge exactly once	Visits every vertex exactly once
Euler circuit: all vertex degrees even in a connected graph	No similarly simple necessary-and-sufficient degree test
Degree-based test is central	Usually requires structural/case analysis

High-value exam traps

- Euler = EDGE; Hamilton = VERTEX.
- A graph with exactly two odd-degree vertices can have an Euler path but not an Euler circuit, assuming connectedness.
- A tree with n vertices has $n-1$ edges.
- A bipartite graph has no odd-length cycle.

EXAM QUESTION BANK — 1, 2, 5 & 8 MARKS

Unit I — Logic, Proofs & Sets

Marks	Question	Answer / writing points
1	Define proposition.	Declarative statement with exactly one truth value: T or F.
2	State De Morgan's laws.	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$; mention truth-table verification.
5	Explain quantifiers and their negation.	Define $\forall$ and $\exists$; give examples; write $\neg \forall x P(x) \equiv \exists x \neg P(x)$ and $\neg \exists x P(x) \equiv \forall x \neg P(x)$.
8	Prove $\sqrt{2}$ is irrational.	Assume $\sqrt{2}=a/b$ in lowest terms; derive a even; then b even; contradiction.

Unit II — Functions, Sequences & Recurrence

Marks	Question	Answer / writing points
1	Define bijective function.	A function that is both injective and surjective.
2	Differentiate domain, codomain and range.	Domain=input set; codomain=target set; range=actual outputs.
5	Find inverse and composition for given functions.	Show substitution/algebra step-by-step; apply inner function first in composition.
8	Solve a second-order recurrence.	Write recurrence, characteristic equation, roots, general solution, then use initial conditions.

EXAM QUESTION BANK — UNIT III & IV

Unit III — Counting & Relations

Marks	Question	Answer / writing points
1	Permutation vs combination?	Permutation: order matters. Combination: order does not matter.
2	State generalized Pigeonhole Principle.	At least $\lceil n/k \rceil$ objects occur in one of k boxes when n objects are distributed.
5	Explain relation properties.	Define reflexive, symmetric, antisymmetric and transitive with one example each.
8	Find transitive closure / prove equivalence relation.	List relation pairs; add required transitive pairs until stable OR verify reflexive+symmetric+transitive separately.

Unit IV — Graph Theory & Trees

Marks	Question	Answer / writing points
1	What is a tree?	Connected, acyclic graph.
2	State Handshaking Theorem.	$\sum \deg(v) = 2 E $; hence number of odd-degree vertices is even.
5	Differentiate Euler and Hamilton paths.	Euler covers every edge once; Hamilton visits every vertex once; compare existence tests.
8	Solve a Dijkstra/MST problem.	Draw graph, initialize distances or sort edges, show every selection/update, then state final path/MST and total weight.

Final exam-writing formula

Definition → **Formula/Rule** → **Diagram/Table** → **Working** → **Final Answer**. For proof questions, number every logical step. For graph numericals, draw the graph first.

LAST-MINUTE REVISION — MUST REMEMBER

Topic	Must-remember point
Logic	Contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$; De Morgan's laws.
Quantifiers	Negating $\forall$ changes it to $\exists$, and vice versa.
Sets	$ P(A) = 2^{ A }$; $ A \times B = A \cdot B $.
Functions	Injective = one-to-one; surjective = onto; bijective = both.
Recurrence	Characteristic equation; distinct roots vs repeated roots.
Counting	Permutation = order matters; combination = order does not.
Pigeonhole	At least $\lceil n/k \rceil$ in one box.
Relations	Equivalence = $R+S+T$; partial order = $R+A+T$.
Graphs	$\sum \deg(v) = 2 E $; odd-degree vertices occur in an even number.
Euler	Every edge exactly once.
Hamilton	Every vertex exactly once.
Dijkstra	Shortest paths with non-negative weights.
Planar	$V-E+F=2$ for connected planar graphs.
Tree	Connected + acyclic; n vertices $\rightarrow n-1$ edges.
MST	Kruskal = edge-focused; Prim = vertex-focused.
Coloring	$\chi(G)$ = minimum number of colors for adjacent vertices to differ.
Traversal	Pre: R-L-R; In: L-R-R; Post: L-R-R — remember as Root position: first/middle/last.

10-minute final revision: formulas $\rightarrow$ definitions $\rightarrow$ common differentiations $\rightarrow$ one recurrence $\rightarrow$ one P&C; $\rightarrow$ one relation closure $\rightarrow$ one Dijkstra $\rightarrow$ one MST $\rightarrow$ tree traversals.

This addendum was prepared specifically to address the improvement points identified by comparing the uploaded notes with the official Semester III DMS syllabus.